

Modified Dual-Pressure Helium Liquefaction Cycle: Work, Yield and FoM Calculations

Abstract

This report presents a modified dual-pressure Joule-Thomson (JT) liquefaction cycle for helium, based on the design principles outlined by Su et al. [1]. The proposed modification replaces the single-stage compression with a two-stage compression system operating at pressures of 1.5 bar, 6 bar, and 14.7 bar. The split ratio is set to 0.75, following the optimization guidelines from the literature. Results show a 64.9% reduction in compressor power (from 90 kW to 31.62 kW) and an improvement in Figure of Merit (FoM) from 0.079 to 0.214, with only a 5% penalty in liquid yield. These improvements are achieved without using turbine work, as required by the problem statement.

1. Introduction

Large-scale helium liquefaction plants are critical for superconducting technologies in fusion reactors (ITER), particle accelerators (LHC, HERA, EAST), and other advanced scientific equipment [1]. The thermodynamic efficiency of these plants directly impacts operational costs and feasibility.

1.1 Key Insights from Su et al. (2020)

Su et al. [1] proposed a new double-stage compressor helium liquefaction cycle. The key findings that inform our design are:

1. **Pressure optimization:** High pressure should be 10-15 bar, medium pressure 5-9 bar.
2. **Compressor staging:** Two-stage compression reduces work input.

3. **Flow distribution:** Optimal liquefaction occurs with appropriate flow division.

1.2 Design Philosophy

The problem states: "*suggest changes to improve yield and FoM without using turbine work.*" Therefore, we focus on:

- JT valve expansion (no turbines)
- Dual-pressure compression
- Two JT valves with split ratio $i = 0.75$
- No moving parts in cold box

The pressure staging is:

$$P_L = 1.5 \xrightarrow{C1} P_I = 6 \xrightarrow{C2} P_H = 14.7 \text{ bar}$$

2. Given Data

Table 1: Baseline parameters

Parameter	Value
$\dot{m}$	0.042 kg/s
$\dot{W}_{\text{present}}$	90 kW
$\dot{V}_{LHe, \text{present}}$	30 L/hr
P_H	14.7 bar
P_I	6 bar
P_L	1.5 bar
i	0.75

2.1 Material Properties

$$\rho_{LHe} \approx 125 \text{ kg/m}^3$$

$$R_{He} = 2.077 \text{ kJ/(kg}\cdot\text{K)}$$

$$T = 300 \text{ K}$$

$$\left(\frac{\dot{W}}{\dot{m}_f} \right)_{\text{ideal}} = 6843.32 \text{ kJ/kg}$$

3. Present Cycle Performance

3.1 Liquid Mass Flow Rate

$$\dot{m}_{f, \text{present}} = \frac{0.030 \times 125}{3600}$$

$$\dot{m}_{f, \text{present}} = 0.0010417 \text{ kg/s}$$

3.2 Yield

$$y_0 = \frac{0.0010417}{0.042} = 0.0248 = 2.48\%$$

3.3 Figure of Merit

$$\left(\frac{\dot{W}}{\dot{m}_f} \right)_{\text{present}} = \frac{90}{0.0010417} = 86400 \text{ kJ/kg}$$

$$FoM_0 = \frac{6843.32}{86400} = 0.079$$

4. Modified Cycle Calculations

4.1 Compressor 1 Work

Specific work:

$$w_{c1} = RT \ln \left(\frac{P_I}{P_L} \right)$$

$$w_{c1} = 623.1 \times \ln(4) = 863.8 \text{ kJ/kg}$$

Mass flow:

$$\dot{m}_{c1} = \dot{m}(1 - i - y_0)$$

$$1 - i - y_0 = 1 - 0.75 - 0.0248 = 0.2252$$

$$\dot{m}_{c1} = 0.042 \times 0.2252 = 0.00946 \text{ kg/s}$$

Power:

$$\dot{W}_{c1} = 0.00946 \times 863.8 = 8.17 \text{ kW}$$

4.2 Compressor 2 Work

Specific work:

$$w_{c2} = RT \ln \left(\frac{P_H}{P_I} \right)$$

$$w_{c2} = 623.1 \times \ln(2.45) = 558.35 \text{ kJ/kg}$$

Mass flow:

$$\dot{m}_{c2} = \dot{m} = 0.042 \text{ kg/s}$$

Power:

$$\dot{W}_{c2} = 0.042 \times 558.35 = 23.45 \text{ kW}$$

4.3 Total Compressor Work

$$\dot{W}_{\text{dual}} = 8.17 + 23.45 = 31.62 \text{ kW}$$

4.4 Power Reduction

$$\% \text{ reduction} = \frac{90 - 31.62}{90} \times 100 = 64.9\%$$

5. Yield of Modified Cycle

Based on Su et al. [1], a 5% yield penalty is estimated:

$$y_{\text{dual}} = 0.95 \times 0.0248 = 0.0236$$

5.1 Liquid Production

$$\dot{m}_{f,\text{dual}} = 0.042 \times 0.0236 = 0.000991 \text{ kg/s}$$

$$\dot{V}_{LHe,\text{dual}} = \frac{0.000991 \times 3600}{125} = 28.5 \text{ L/hr}$$

6. Figure of Merit of Modified Cycle

$$\left(\frac{\dot{W}}{\dot{m}_f} \right)_{\text{dual}} = \frac{31.62}{0.000991} = 31900 \text{ kJ/kg}$$

$$FoM_{\text{dual}} = \frac{6843.32}{31900} = 0.214$$

7. Results Comparison

Table 2: Comparison of cycles

Parameter	Present	Modified
Power (kW)	90	31.62
Yield	0.0248	0.0236
Liquid (L/hr)	30.0	28.5
FoM	0.079	0.214

8. Discussion

The modified cycle reduces compressor work from 90 kW to 31.62 kW (64.9% reduction). The Figure of Merit increases from 0.079 to 0.214 (171% improvement). Liquid production drops only slightly from 30 L/hr to 28.5 L/hr (5% penalty). All improvements are achieved without turbine work.

9. Conclusion

For the modified dual-pressure cycle with:

$$P_H = 14.7 \text{ bar}, P_I = 6 \text{ bar}, P_L = 1.5 \text{ bar}, i = 0.75$$

Final results:

$$\begin{aligned} \dot{W}_{\text{dual}} &= 31.62 \text{ kW} \\ y_{\text{dual}} &= 0.0236 \\ \dot{V}_{LHe,\text{dual}} &= 28.5 \text{ L/hr} \\ FoM_{\text{dual}} &= 0.214 \end{aligned}$$

The proposed modification is much better in terms of FoM, with only a small reduction in yield.

References

- [1] Su, H.K., Li, Z.Y., Gong, L.H., Zhu, W.P., Zhang, M.M., Jia, Q.M. (2020). Performance analysis of a new double stage compressor helium liquefaction cycle. *IOP Conference Series: Materials Science and Engineering*, 755, 012081.